Supporting Information

February 22, 2026

S.1 Methods

S.1.1 The PIGEAN model

Given a trait (which may be either a quantitative trait or liability for disease), we assume that each gene $i \dots N$ either is or is not involved in the trait, indicated as $d_i = 1$ and $d_i = 0$ respectively. We refer to genes with $d_i = 1$ as trait-relevant (or disease-relevant); these can be interpreted as genes that can be perturbed in a human (i.e. via a compound or editing of the gene sequence) and result in an altered value for the trait. This binary notion of trait-relevance is a simplification of reality and does not capture the notion that some genes may permit perturbations that cause larger effects on the trait than other genes.

By definition, trait-relevant genes are a superset of the genes for which genetic variants will exhibit association with the trait. However, many trait-relevant genes will not have variants with reported associations, due to either (a) insufficient power of current datasets to detect association between the trait and a variant that impacts the gene's coding sequence or regulation; or (b) a lack of variants in the population (either due to evolutionary forces or luck) that perturb the gene in a manner that affects the trait. Therefore, although our goal is ultimately to elucidate the full set of genes relevant to the trait, human genetics will produce an incomplete list due to either transient data limitations or fundamental population limitations.

We use $\mathcal{D}_i$ to refer to association data for gene i. We consider four types of association data:

- GWAS association data consists of common variants linked to the gene and their summary statistics (p-values, effect sizes, and standard errors). GWAS summary statistics can be downloaded from resources such as the GWAS catalog[] or the Association to Function Knowledge Portal[].

- "Known" disease gene lists consist of a set of genes previously established as harboring variants that influence the trait, together with a confidence level (represented as a probability) that the gene is in fact relevant to the rare disease. Disease gene lists for rare diseases are available (with confidence levels) from resources such as Orphanet[] or the GenCC[], while gene lists for complex traits are available from resources such as the Human Phenotype Ontology[].

- Rare disease exome sequence data consists of case and control variant counts and frequencies, together with annotations of variant deleteriousness (i.e. from REVEL[], AlphaMissense[], or LofTee[]).

- Complex trait exome sequence data consists of gene-level summary statistics (p-values, effect sizes, and standard errors), obtained from burden tests across one or more masks and collapsed into a single aggregate score for each gene.

In the current manuscript, we focus only on GWAS association data and known disease gene lists, but we derive equations for all four cases below.

We define the "direct genetic support" of a gene as $\Pr(d_i|\mathcal{D}_i)$ – that is, the probability that the gene is trait-relevant conditional upon the association data observed for it. If multiple types of genetic association data is available for the gene, then the direct genetic support is conditional upon all types of association data; i.e. we can explicitly define

$$\Pr(d_i|\mathcal{D}_i) = \Pr\left(d_i|\mathcal{D}_i^g, \mathcal{D}_i^k, \mathcal{D}_i^c, \mathcal{D}_i^e\right)$$

where $\mathcal{D}_i^g$, $\mathcal{D}_i^k$, $\mathcal{D}_i^c$, and $\mathcal{D}_i^e$ are GWAS, known genes, rare disease exome sequencing counts, and complex trait exome gene-level associations.

We define the "indirect genetic support" of a gene as $\Pr(d_i|\mathcal{D}_{-i})$, where $\mathcal{D}_{-i} := \{\mathcal{D}_k : k \neq i\}$ – that is, the probability that the gene is trait-relevant conditional upon the association data observed for all genes other than i (for GWAS association data, it is possible that association data for genes i and k may share SNPs in common if the genes are nearby each other and SNPs have probabilities of being linked to multiple genes). The indirect genetic support of a gene combines with its direct support to produce its combined support, or $\Pr(d_i|\mathcal{D}_i, \mathcal{D}_{-i}) = \Pr(d_i|\mathcal{D})$ The intuition as to why association data for gene k can influence the genetic support for gene i is if the two genes are in a common biological process – if the process is relevant to the trait, as demonstrated by an association observed for variants of gene k, then it increases the probability that other genes in the process are also relevant to the trait. Although it is not necessary for the subsequent derivation, it a biological process can be thought of mathematically as a (sparse) function that maps genes to probabilities of membership in the process; or, equivalently, a set of genes annotated with their probability of membership in the process.

We do not observe biological processes (in the language of machine learning, they are "latent" structures), but there are many experimental or *ad hoc* approaches that produce noisy readouts of processes. For example,

- databases such as Reactome[], Wiki processes[], or the Gene Ontology[] represent curated knowledge of biological pathways and gene functional categories
- genes whose expression is up or down regulated in response to a cellular stimulus are more likely to participate in a common process
- genes whose alteration causes a similar animal model phenotype are more likely to participate in a common process
- genes that co-occur in the literature with a biological concept (or term) are more likely to participate in a common process

These readouts can be modeled as gene "functional annotations" – that is, any numerical value that can be assigned to genes. An important special case of a functional annotation is a gene set – this is the case when genes are assigned either values of zero or one. However, a functional annotation is a more general term and can encompass values such as expression levels in a tissue or co-occurrence weights with a literature term. There are millions of functional annotations available for genes[], from varying sources and of varying quality. Scientists use functional annotations constantly and often implicitly to prioritize genes for further study. The challenge is that with so functional annotations available, it is unclear which ones are most predictive of trait-relevant processes, the degree to which they predict trait-relevant processes, and the extent to which they are additive as opposed to redundant.

We model the degree to which each annotation predicts trait-relevance through the following equations:

$$d_i \sim \text{Bern}\left(y_i\right)$$

$$\log(\frac{y_i}{1-y_i}) = \sum_j X_{ij}\beta_j + \alpha_i + \log\left(\frac{\pi}{1-\pi}\right)$$

$$\beta_j \sim \delta_j \mathcal{N}\left(0, \sigma_j^2\right)$$

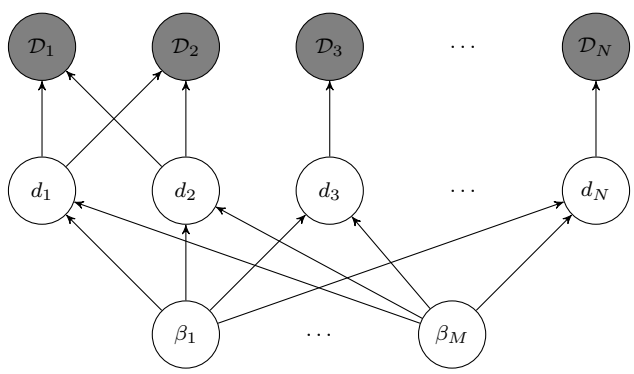
$$\alpha_i = \alpha \sum_j X_{ij}$$

$$\sigma_j^2 = \sigma^2 \text{sd}_i\left(X_{ij}\right)^w$$

$$\delta_j \sim \text{Bern}(p)$$

Where X_{ij} for $j = 1 \ldots M$ is the value of annotation j for gene g_i, β_j is the "effect size" of annotation j on the gene's (log-odds) probability of disease relevance, π is the background (or average) probability of disease relevance, and α, σ, p, and w are regularization parameters. Note that the d_i and β_j values are latent, while the X_{ij} values are fully observed.

A graphical model of these random variables is shown in Figure S.1.1.



The regularization parameters control model sparsity and over-fitting. Specifically:

- We impose a "spike and slab" prior on the β_j values, which asserts that (a) most annotations have zero effect on a gene's trait-relevance and (b) those that do have effects on the (log-odds) of trait-relevance that are normally distributed. The parameter p controls the probability that the effect is non-zero, and the σ_j parameter controls the variance of the conditional effect size distribution

- We use an annotation specific variance σ_j related to a power w of the variance of the gene annotation (across genes). Intuitively, "larger" or more profligate annotations (i.e. those with more genes having non-zero values) should have lower power to predict trait-relevance, since they are less specific biologically. Traditional gene set enrichment approaches implicitly assume that $w = 0$, which biases results toward larger gene sets, as those have smaller standard errors. A value of $w = -2$ adjusts for this bias, and is commonly used in the mathematically similar scenario of modeling SNP effect sizes[].

- Analogous to the w parameter, we use an α parameter to control biases in favor of genes with more annotations. For reasons discussed below**??**, genetic support is biased toward positive values, and therefore genes with more annotations (i.e. better studied genes) typically have higher support unless this is controlled for. We typically set α (i.e. via linear regression) such that there is no relationship between a gene's size and its d_i value.

S.1.2 Fitting the model

Because the d_i and β_j values are latent, we must infer them from the observed association data $\mathcal{D}$ across all genes. Specifically, we want to infer the joint distribution

$$\Pr(d, \beta | \mathcal{D})$$

Analytically determining an equation for this distribution is intractable, and therefore we use Gibbs sampling to sample from the joint distribution. Gibbs sampling proceeds by iteratively drawing samples from the distribution of each variable conditional upon the current samples for all other variables. In our case, we have N variables d_i and M variables β_j. A basic Gibbs sampler would need to derive and sample from the conditional distributions

$$\Pr(d_i | \mathcal{D}, d_{-i}, \beta)$$

$$\Pr(\beta_j | d, \mathcal{D}, \beta_{-j})$$

Instead, we employ a more efficient form of Gibbs sampling, termed block Gibbs sampling, that iterates between sampling all d values and all β values simultaneously:

$$\Pr(d | \mathcal{D}, \beta)$$

$$\Pr(\beta | d, \mathcal{D}) = \Pr(\beta | d)$$

The right hand side of the second equation follow from the conditional dependencies encoded in our model, implying that we can ignore association data when sampling from the distribution for β if the d values are observed, as in this case the association data adds no additional information.

Sampling from $\Pr(\beta | d)$ is itself non-trivial. The equations for the posterior distribution of variables with spike and slab priors has previously been studied in the context of polygenic risk scores – specifically, the LD-pred method[]. In this work, the true effect sizes of SNPs on a quantitative trait have a spike and slab prior, they affect the trait additively, and the observed data is the marginal estimates of the SNP effect sizes from a series of univariate regressions of observed trait values (in individuals) and the value (0, 1, or 2) of the SNP. Furthermore, the equations used to derive the posterior estimates of SNP effect sizes in this setting have been shown to apply to binary traits as well, provided the marginal SNP effect sizes are expressed in units of $\log(OR)$[], which holds if they are obtained from a logistic regression of individual disease status on SNP value. In our setting, the equations relating annotation effect sizes β_j to gene trait-relevance d_i are identical to those relating SNP effect sizes to individual disease status (intuitively, annotations correspond to SNPs and genes correspond to individuals). Therefore, we can follow the same procedure used by LD-pred[] to sample from the joint distribution of β values.

Practically, this entails (a) running a univarate logistic regression of d on each β_j to obtain marginal annotation effect size estimates (as well as standard errors); and (b) running a Gibbs sampling procedure to convert the marginal annotation effect size estimates into posterior joint effect estimates – that is, both correcting for correlations across annotations and accounting for the spike and slab prior. The spike and slab prior at this stage plays the role of regulariztion, both

shrinking annotation effect sizes toward zero and enforcing sparsity. The Gibbs sampler for the β_j has been described previously, and at a high-level entails (a) iterating over each annotation j; (b) subtracting the current β_{-j} values from its marginal effect size estimate, proportional to the degree of correlation between the other annotations and annotation j; and then (c) applying a spike and slab shrinkage equation to the residual value of β_j. This procedure is iterated until the β values converge (in practice we run four chains in parallel to evaluate convergence[]).

The equations for sampling from $\Pr(d|\mathcal{D},\beta)$ depend on the type of association data that is present. For GWAS association data, the calculations are complex and we address those below. For known disease gene lists, rare disease exome count data, and complex trait gene-level associations, association data for different genes is independent. The sampling equations then reduce to:

$$\begin{aligned}\Pr(d|\mathcal{D},\beta) &= \prod_i \Pr(d_i|\mathcal{D},\beta) \\ &= \prod_i \Pr(d_i|\mathcal{D}_i,\beta)\end{aligned}$$

where $\mathcal{D}_i$ is the set of variants in the association data linked to gene i. Since d_i is a binary variable, we can encode this conditional distribution in terms of its posterior odds; namely

$$\Pr(d_i|\mathcal{D}_i,\beta) = \frac{PO_i}{(1+PO_i)}$$

where

$$\begin{aligned}PO_i &= \frac{\Pr(d_i=1|\mathcal{D}_i,\beta)}{\Pr(d_i=0|\mathcal{D}_i,\beta)} \\ &= \frac{\Pr(d_i=1|\mathcal{D}_i,\beta)}{1-\Pr(d_i=1|\mathcal{D}_i,\beta)}\end{aligned}$$

In turn, PO_i can be expressed as

$$PO_i = BF_i \frac{\Pr(d_i=1|\beta)}{1-\Pr(d_i=1|\beta)}$$

where BF_i is the Bayes Factor for d_i:

$$\begin{aligned}BF_i &= \frac{\Pr(\mathcal{D}_i|d_i=1,\beta)}{\Pr(\mathcal{D}_i|d_i=0,\beta)} \\ &= \frac{\Pr(\mathcal{D}_i|d_i=1)}{\Pr(\mathcal{D}_i|d_i=0)}\end{aligned}$$

where the second equality follows from the conditional independence assumptions of our model.

For known disease gene lists and exome sequence data, the BF_i depend only on the association data and can therefore be computed at the beginning of running the PIGEAN model.

- For known disease gene lists, we are given a confidence value c_i for each gene (or we can use a fixed appropriate constant such as 95% or 99%), and can use $BF_i = \frac{c_i}{1-c_i}$
- For rare disease exome sequence counts, we can compute bayes factors for each gene following the exTADA model[].

- For complex trait exome sequence gene-level associations, we can calculate Bayes Factors using the same approach as articulated in the HuGE method[]. Briefly, gene-level effect sizes and standard errors are converted to Bayes Factors using an approach previously formulated for SNP associations, leveraging the similarity between a single-SNP association model and a burden test association model (a burden test in effect collapses all rare variants in a gene into an "uber-SNP").

The second needed quantity is given directly by our model as

$$\frac{\Pr\left(d_i = 1|\beta\right)}{1 - \Pr\left(d_i = 1|\beta\right)} = \sum_j X_{ij}\beta_j + \alpha_i + \log\left(\frac{\pi}{1-\pi}\right) \tag{S1}$$

Therefore, in summary, to sample from $\Pr\left(d_i|\mathcal{D}_i, \beta\right)$ in the case where the gene-level association data is independent across genes and can be summarized as Bayes Factors in advance, we simply multiply the Bayes Factors by a "prior" for the gene that is the product of the initial prior (π) and a Bayes Factor derived from the current β values and the gene's annotations. This equation also emphasizes an equivalent interpretation of indirect genetic support: it can be viewed as an informed prior for each gene, based on the gene's annotations and calibrated according to genetic association data for all genes in the genome.

Sampling from the $\Pr\left(d|\mathcal{D}, \beta\right)$ distribution in the case of GWAS association data is completely analogous, except (as we discuss below) the BF_i values may change as a function of the β values. This is because even when the β are observed, the gene-level d_i values can influence each other through shared association data (since the association data is observed).

Finally, in the case when there are multiple types of association data (i.e. a trait with GWAS data, exome sequence associations, and a known gene list), provided that the variants included in the association evidence are independent (i.e. not in linkage disequilibrium), the Bayes Factors for the genes simply multiply. When calculating the GWAS Bayes Factors at each iterations, the Bayes Factors for the other types of association data should multiply the informed prior used in their calculation (see below for how this is done).

These equations are sufficient to define our Gibbs sampling procedure. It consists of an outer sampling loop and an inner sampling loop. The outer sampling loop draws d values from the current β values. The inner sampling loop calculates the β values given the current d values. In each case we run 4 chains and evaluate convergence using a widely used approach[]. For the inner loop, we return a sample of β immediately after convergence. For the outer loop, we run additional iterations to get sufficient resolution as to the mean of d and β values. Before calculating the final mean β and d values, we remove any chains whose mean is above 10 median absolute deviations from the mean across chains.

S.1.3 Human Genetic Evidence Refined (HuGER) scores

To calculate $\Pr\left(d|\mathcal{D}, \beta\right)$ when the association data $\mathcal{D}$ is from GWAS SNPs, one would ideally implement a rigorous probabilistic assignment of gene-level probabilities that account for various aspects of the "variant to function"[] problem, including the probability that a GWAS association exists nearby the gene, SNP-level fine mapping to determine SNP causality (in the face of LD) in the event that one does, epigenomic annotations that affect the probability of SNP causality during fine mapping, predictions of SNP to gene linkages, and the proximity of other genes nearby the gene that might "compete" for linkage to the causal SNP. Such a "gene-level fine mapping" approach has been studied in the context of eQTLs[] and transcriptome-wide association studies[],

and extending these ideas to the general case of probabilistic gene-level fine mapping is a potentially valuable research direction.

We instead proceed by building upon a previously proposed method for calculating gene-level Bayes Factors from GWAS – HuGE scores[]. In their original formulation, HuGE scores are based on a set of simple rules for assigning probabilities to genes based on (a) whether a genome-wide significant association lies within a window of the gene, (b) whether a genome-wide significant association is in the coding sequence of the gene, (c) whether the gene is the nearest gene to the association, and (d) whether the variant with the lowest-p-value in the region is in the coding sequence of the gene. The probabilities are determined by empirical observations (e.g., that the nearest gene is the effector gene for a GWAS association roughly 70% of the time[]).

Here, we extend the HuGE framework in four ways, addressing its key limitations while retaining its simplicity, interpretability, and low computational cost:

1. Rather than assigning a gene a positive HuGE score if it is within a fixed window of a GWAS association, we continuously adjust the HuGE score according to how likely it is to be linked to the causal SNP for the GWAS association, either using known S2G[] or V2G[] SNP to gene probabilities or a continuous function calibrated such that the nearest gene to a causal SNP is the effector 70% of the time[]

2. Rather than assigning HuGE scores independently to every gene within a window of GWAS association, we "divide" the HuGE score among genes linked to the GWAS association, according to their probability of linkage to the association and their priors.

3. Rather than considering only the lead SNP of a GWAS association, we fine map the association across variants, assigning each a probability of causality, and then take the expectation of HuGE scores over potential causal variants.

4. Rather than considering only genome-wide significant SNPs when calculating HuGE scores, we consider all SNPs in the genome, as even SNPs below genome-wide significance have a non-zero probability of being causally associated with the trait.

We first describe how the method works at a high level before deriving equations to justify the approach. Given a file of GWAS summary statistics (variant p-values, effects sizes, standard errors, and/or effective sample sizes):

1. We identify the strongest associated SNP remaining in the summary statistics; this SNP represents a GWAS signal.

2. We calculate the probability that the GWAS signal is truly associated, as a function of its p-value, using a previously derived equation[**?**].

3. We expand this lead SNP into a window likely to contain all SNPs in LD with it. This could be done with an LD-clumping. approach[], but to keep the dependencies of running HuGER scores minimal, we perform an approximate clumping approach that extends the window until the SNP p-values either rise above some fraction of that of lead SNP, or until the SNP p-values begin to decline again (indicating a potential independent signal).

4. We apply approximate fine mapping to convert the SNP p-values to a credible set of variants, each with a posterior probability of association[], using a simple approach that does not require LD or approximate conditional analysis.

5. For each variant in the credible set, we probabilistically link it to each gene (the sum of link probabilities across all genes equals 1). If S2G[] or V2G[] scores are input, these are used as probabilities; otherwise we fit a logistic decay function tuned such that (for SNPs randomly sampled from the summary statistics file) the nearest gene to the SNP is linked to the SNP 70% of the time. In reality, a Weibull function rather than a logistic function is probably a better link function.

6. We adjust the variant to gene link probabilities for each gene's prior value (assuming the priors are non-uniform, which is the case if we are running PIGEAN with non-zero β values). Details of this procedure are discussed below.

7. We calculate, for each gene, the probability that it is trait-relevant based on this GWAS signal. This probability is the probability that two events are true: the signal is truly associated, and the gene is linked to the causal variant for the signal. We calculate this based on the overall probability that the signal is truly associated, the variant posterior probabilities, and the variant to gene linkage probabilities.

8. We repeat these steps until there are no SNPs left in the genome.

9. We convert the gene probabilities according to each signal into Bayes Factors (accounting for the gene's prior), then multiply these across signals to obtain an aggregate Bayes Factor, and convert this into a final posterior probability.

The mathematical details of this procedure are as follows. Our goal is to calculate

$$\Pr\left(d|\mathcal{D},\beta\right)$$

where $\mathcal{D}$ is a set of association summary statistics across SNPs in the genome. We first decompose these data into a series of sets $\mathcal{D}^k, k = 1..d$ that consist of independent groups of SNPs; practically, we achieve this by (as discussed above) iteratively choosing the SNP with the lowest p-value, adding it and all SNPs in LD with it as a new dataset $\mathcal{D}^k$, and continuing until all SNPs are exhausted. Assuming that the $\mathcal{D}^k$ are independent (an approximation, since in general some amount of LD will still span SNP clumps), and defining $s(k)$ as the genes linked to SNPs in $\mathcal{D}^k$, we have

$$\begin{aligned}
\Pr\left(d|\mathcal{D},\beta\right) &= \Pr\left(d|\mathcal{D}^1,\ldots,\mathcal{D}^d,\beta\right) \\
&\propto \frac{\prod_k \Pr\left(d|\mathcal{D}^k,\beta\right)}{\Pr\left(d|\beta\right)} \\
&= \frac{\prod_k \Pr\left(d_{s(k)},d_{-s(k)}|\mathcal{D}^k,\beta\right)}{\Pr\left(d|\beta\right)} \\
&= \frac{\prod_k \Pr\left(d_{s(k)}|\mathcal{D}^k,\beta\right)\Pr\left(d_{-s(k)}|\mathcal{D}^k,\beta\right)}{\Pr\left(d|\beta\right)} \\
&= \frac{\prod_k \Pr\left(d_{s(k)}|\mathcal{D}^k,\beta\right)\Pr\left(d_{-s(k)}|\beta\right)}{\Pr\left(d|\beta\right)} \\
&= \prod_k \frac{\Pr\left(d_{s(k)}|\mathcal{D}^k,\beta\right)}{\Pr\left(d_{s(k)}|\beta\right)}
\end{aligned}$$

where the proportionality follows from Bayes' rule and an assumption of conditional independence of the $\mathcal{D}^k$ given the d. The fourth rearrangment follows from the conditional independence assumptions

of our model (conditional upon β, d_i values can only influence each other through shared association data). Our task therefore simplifies to computing, for each independent set of variants $\mathcal{D}^k$ and corresponding genes $s(k)$ linked to SNPs within $\mathcal{D}^k$,

$$\Pr\left(d_{s(k)}|\mathcal{D}^k,\beta\right)$$

We first partition this event into two events: that a disease-associated SNP lies within $\mathcal{D}^k$ (event S) and is mediated by one of the genes in $s(k)$, and the complement of this event (event $\neg S$). Since these events are disjoint,

$$\begin{aligned}
\Pr\left(d_{s(k)}|\mathcal{D}^k,\beta\right) &= \Pr\left(d_{s(k)},S|\mathcal{D}^k,\beta\right)+\Pr\left(d_{s(k)},\neg S|\mathcal{D}^k,\beta\right)\\
&= \Pr\left(d_s(k)|S,\mathcal{D}^k,\beta\right)\Pr\left(S|\mathcal{D}^k,\beta\right)\\
&\quad +\Pr\left(d_{s(k)}|\neg S,\mathcal{D}^k,\beta\right)\Pr\left(\neg S|\mathcal{D}^k,\beta\right)\\
&\approx \Pr\left(d_{s(k)}|S,\mathcal{D}^k,\beta\right)\Pr\left(S|\mathcal{D}^k\right)+\Pr\left(d_{s(k)}|\neg S,\mathcal{D}^k,\beta\right)\Pr\left(\neg S|\mathcal{D}^k\right)
\end{aligned}$$

where the approximate equality assumes that, given the summary statistics, the probability of a disease-associated SNP lying in the region is not further affected by knowledge of gene causal status or its annotations (note that this would clearly not hold if the summary statistics were not observed).

We first calculate $\Pr\left(S|\mathcal{D}^k\right)$ as the probability that the strongest SNP in the region is truly associated with the trait, using a previously derived equation for converting association Z-scores to Approximate Bayes Factors[] – specifically,

$$ABF^k=\sqrt{1+K}\exp\left(-\frac{z^2}{2}\frac{K}{(1+K)}\right)$$

where z is the Z-score of the the strongest SNP association in $\mathcal{D}^{\|}$ and K is a free parameter. Additionally, to convert these Bayes Factors to a posterior probability, we need to specify the prior probability ϕ that a SNP is truly associated with the trait, which introduces an additional free parameter. We choose these parameters by specifying two Z-score values (equivalenetly: two p-value thresholds) and their corresponding posterior probabilities. These are more intuitive ways of specifying these parameters and are inputs to PIGEAN. Default values are a probability of 0.75 for $p=5\times10^{-8}$ and a probability of 0.01 for $p=0.01$. However, it is known[] that this equation for converting Z-scores to ABF values implicitly depends on the sample size of the GWAS; specifically, its prior over SNP effect sizes has a stronger belief in smaller effect sizes as the sample size increases. While corrections are possible to this equation that remove this dependence (as are corrections that use effect sizes and standard errors instead), we believed that these corrections would require users of PIGEAN to specify too many non-intuitive parameters and impose difficulty in running PIGEAN across many traits for which these parameters might vary. Therefore, we instead imposed an approximate sample size correction where, based on the number of signals that reach genome-wide significance, we adjust the mapping between p-value and ABF; specifically, we empirically reset the p-value threshold at which the posterior equals 0.75 to the p-value of the 100th strongest signal (rather than fixing it at $p=5\times10^{-8}$). We found empirically that this worked well across many traits. It is interesting (if perhaps underappreciated) that these same considerations motivate a more stringent genome-wide significance threshold for GWAS as sample sizes increase, which to our knowledge is not standard practice in the genetics community.

Combined, we have

$$\Pr\left(S|\mathcal{D}^k\right) = \frac{ABF^k * \frac{\phi}{1-\phi}}{1 + ABF^k * \frac{\phi}{1-\phi}}$$

The other two quantities that we need to calculate are

$$\Pr\left(d_{s(k)}|S, \mathcal{D}^k, \beta\right)$$

$$\Pr\left(d_{s(k)}|\neg S, \mathcal{D}^k, \beta\right)$$

For the second quantity, we assume that if there are no significant SNPs, then

$$\Pr\left(d_{s(k)}|\neg S, \mathcal{D}^k, \beta\right) = \Pr\left(d_{s(k)}|\beta\right)$$

or the prior probability for the gene (given the β values), which can be calculated as in equation (S1).

For the first quantity, we assume that exactly one gene is causal in the region. This assumption is equivalent to assuming that SNPs exert effects through at most one effector gene, as conditioning on the event S implies that there is at least one causal gene in the region. This assumption likely does not hold for all regions of the genome, but it probably holds for many regions of the genome. It is a limitation of the HuGER model that could be addressed in a more comprehensive gene "fine mapping" approach as discussed above.

In the context of credible set calculation for SNPs, but under these same assumptions (exactly one causal SNP in a set), it has been shown**??** that the posterior probability that the causal gene is gene i is proportional to its Bayes Factor multiplied by the conditional prior of d_i – that is, the prior probability of d_i before any SNP statistics are observed but after it is known that exactly one of the genes in the region is causal. Specifically:

$$\Pr\left(d_i|S, \mathcal{D}^k, \beta\right) \quad = \quad \alpha BF_i \Pr\left(d_i|\oplus_{j\in s(k)}, \beta\right)$$

where we choose α such that the resulting values to sum to 1 across all genes in the region of the signal. The notation $\oplus_{j\in s(k)}$ indicates the event that exactly one gene linked to SNPs in $\mathcal{D}^k$ is causal, and the bayes factor BF_i incorporates all information contained within $\mathcal{D}^k$. Because of our assumption that exactly one gene in the region is causal, calculating $\Pr\left(d_i|S, \mathcal{D}^k, \beta\right)$ for each gene $i \in s(k)$ is sufficient to calculate the full joint probability of $\Pr\left(d|S, \mathcal{D}^k, \beta\right)$.

The event $\oplus_{j\in s(k)}$ is equivalent to $\cup_{i\in s(k)} d_i \cap_{j\in -i} \neg d_j$, or the union over events in which one gene is causal and the rest are not. We can therefore calculate $\Pr\left(d_i|\oplus_{j\in s(i)}, \beta\right)$ by calculating $\Pr(d_i, \cap_j \neg d_j|\beta)$ and then normalizing these values so that they sum to 1. We can approximate this as $\Pr(d_i|\beta)\prod_j (1 - \Pr(d_j|\beta))$ under the simplifying assumption that the priors are independent, and we can calculate each value $\Pr(d_i|\beta)$ as in equation (S1).

To calculate BF_i, we would traditionally model $\Pr\left(\mathcal{D}^k|d_i\right)$ and $\Pr\left(\mathcal{D}^k|\neg d_i\right)$ and then calculate their ratio. However, modeling these quantities is non-trivial and warrants its own full investigation, as described above. We therefore proceed by directly estimating $\Pr\left(d_i|\mathcal{D}^k\right)$, using ideas from the original HuGE calculator[], and then divide the resulting odds by the non-informative prior odds (i.e. not conditional upon β, since we are not conditioning on β in this scenario). That is,

$$\begin{aligned}\frac{\Pr\left(d_i|\mathcal{D}^k\right)}{\Pr\left(\neg d_i|\mathcal{D}^k\right)} &= \frac{\Pr\left(\mathcal{D}^k|d_i\right)}{\Pr\left(\mathcal{D}^k|\neg d_i\right)}\frac{\pi}{1-\pi} \\ &= BF_i\frac{\pi}{1-\pi}\end{aligned}$$

And thus

$$BF_i = \frac{\Pr\left(d_i|\mathcal{D}^k\right)}{\Pr\left(\neg d_i|\mathcal{D}^k\right)}\frac{1-\pi}{\pi}$$

The reason for this seemingly non-intuitive and indirect way of obtaining gene-level Bayes Factors is that, while modeling $\Pr\left(\mathcal{D}^k|d_i\right)$ is difficult, it is more straightforward to obtain an estimate of $\Pr\left(d_i|\mathcal{D}^k\right)$ under the assumption of a single causal variant within $\mathcal{D}^k$.

Specifically, we first obtain a probability distribution $p(V)$, where V indicates the event that variant $v \in \mathcal{D}^k$ is the causal variant within $\mathcal{D}^k$. We can obtain $p(V)$ via any fine mapping approach that produces posterior probabilities of association (within PIGEAN, we use a simple approach that does not require LD estimates or conditional analysis[]). We then combine these with SNP to gene linkage probabilities $p(L_i^v)$, where L_i^v is the event that variant v is linked to gene i and we assume every causal variant is linked to exactly one gene (and thus $p(L_i^v)$ sums to 1 across all genes for each causal variant v). We can obtain these values either directly from external sources or calculate these as a function of distance as described above. The $p(V)$ and $p(L_i^v)$ values then combine as

$$\begin{aligned}
\Pr\left(d_i = 1|\mathcal{D}^k\right) &= \Pr\left(\bigcup_v d_i = 1, V|\mathcal{D}^k\right) \\
&= \Pr\left(\bigcup_v d_i = 1, V, L_i^v|\mathcal{D}^k\right) \\
&= \sum_v \Pr\left(d_i = 1, V, L_i^v|\mathcal{D}^k\right) \\
&= \sum_v p(V)p(L_i^v)
\end{aligned}$$

where the second equality follows because $d_i = 0$ if gene i is not linked to the causal variant, and the last equality follows because of the assumptions that there is exactly one causal variant, it is linked to exactly one gene, and only variants linked to trait-relevant genes can exhibit associations.

Putting all of these steps together, the final HuGER score procedure for each independent group of SNPs $\mathcal{D}^k$ is:

1. For each gene, we calculate the gene-level Bayes Factor

$$\sum_{v\in\mathcal{D}^k} p(V)p(L_i^v)\frac{1-\pi}{\pi}$$

2. We use these Bayes Factors to calculate the full joint distribution over $d_{s(k)}$ values for each $\mathcal{D}^k$.

$$\begin{aligned}
&\Pr\left(d_i|\mathcal{D}^k, \beta\right) = \\
&\quad \Pr\left(S|\mathcal{D}^k\right) \alpha \frac{\sum_{v\in\mathcal{D}^k} p(V)p(L_i^v)}{1-\sum_{v\in\mathcal{D}^k} p(V)p(L_i^v)}\frac{1-\pi}{\pi}\Pr(d_i|\beta)\prod_{j\in s(k)-i}(1-\Pr(d_j|\beta)) \\
&\quad + \left(1-\Pr\left(S|\mathcal{D}^k\right)\right)\Pr\left(d_i|\,\beta\right) \quad \text{(S2)}
\end{aligned}$$

We can then combine these scores to get the full distribution over all d using Equation (S2).

In practice, since enumerating the full joint distribution across d is intractable, we marginalize the values across genes within each $\mathcal{D}^k$ and then combine the marginals across the the $\mathcal{D}^k$. That is, applying Equation (S2) in the case of a single gene, we obtain

$$\frac{\Pr(d_i = 1|\mathcal{D}, \beta)}{\Pr(d_i = 0|\mathcal{D}, \beta)} = \prod_k \frac{\Pr\left(d_i = 1|\mathcal{D}^k, \beta\right)}{\Pr(d_i = 0|\mathcal{D}^k, \beta)} \frac{\Pr(d_i = 1|\beta)}{\Pr(d_i = 0|\beta)}$$

This in effect marginalizes over genes before combining data across $\mathcal{D}^k$, as opposed to combining data across $\mathcal{D}^k$ before marginalizing over genes. We expect this approximation to be reasonable since most genes are linked to SNPs in a small number of $\mathcal{D}^k$, and most $\mathcal{D}^k$ have SNPs linked to only a small number of genes.

S.1.4 Indirect support as a probabilistic justification for previous *ad hoc* approaches

Our definition of indirect genetic support, $\Pr(d_i|\mathcal{D}_{-i})$, demonstrates that association data for every SNP in the genome contributes to the genetic support for a gene. With this formalism in place, it is possible to interpret several previous statistical approaches or ideas as special cases of this concept, albeit often with hueristics or approximations to the full probabilistic calculation.

S.1.5 Investigating suggestive signals

It is common in genetic association studies to report (in addition to signals that exceed genome-wide significance) signals that are "suggestive" and fall just short of study-wide significance. If these signals can be linked to genes previously suggested to play a role in disease, they are often emphasized in a discussion of the study's findings, albeit with no statistical justification. Our model provides statistical guidelines for when such genes can be justifiably reported: the previous evidence implicating a gene in disease can be expressed as indirect genetic support, and if this (when added to the suggestive direct support) exceeds a false discovery threshold, then the gene can be considered to show a significant association. Equivalently, the indirect support value for the gene re-weights its association accounting for an informed gene prior, but because indirect support is estimated from genetic data, there is need for subjective or *ad hoc* prior specification.

S.1.5.1 Prioritizing genes within GWAS loci

There is much interest in prioritizing "effector genes" nearby GWAS signals[**?**]. One widely used computational approach is the PoPS method[], which (a) uses MAGMA[] to calculate gene-level Z-scores from the GWAS; (b) fits a multi-variate ridge regression to predict MAGMA scores as a function of gene annotations; (c) applies this method to each chromosome separately, using only data from the other chromosomes, and obtains PoPS scores based on the model's predictions; and (d) "prioritizes" genes within GWAS loci that have the highest PoPS scores. PoPS scores have no quantitative meaning and cannot be compared across loci or combined with MAGMA scores, which limits their utility outside of effector gene prioritization.

However, in the context of indirect genetic support, it is possible to interpret the assumptions and approximations made by PoPS. Assume we amend our model to (a) compute gene-level Bayes Factors as equivalent to MAGMA Z-scores; (b) impose a normal prior (rather than a spike and slab prior) on the β values; (c) when estimating the β and d_i values, treat d_i values for genes on other chromosomes as observed rather than latent; (d) omit the regularization parameters penalizing well-known genes and large gene sets; and (e) conduct only a single iteration of Gibbs sampling,

using the expected value of the d and β values. Under these amendments, the scores produced by our model are identical to those produced by PoPS. Therefore, PoPS can be viewed as an approximate method for prioritizing genes at each GWAS locus with the strongest indirect genetic support, under the specific assumption that MAGMA Z-scores represent gene-level Bayes Factors.

S.1.5.2 Filtering known versus novel genes

After obtaining the results of a genetic association study for a trait, one of the first tasks is to examine positive controls (do "known" genes exhibit association) and to see if any signals are novel. This typically involves manually compiling lists of known genes and searching the literature to see if significant associations have reported links to the trait. Indirect support provides an automated starting point to conduct these analyses – genes with high indirect support values can be considered as "known" while genes with no indirect support can be considered as novel (or, at minimum, having no annotations shared with other known genes). Often, the most interesting genes that emerge from a study are those that have not been definitely implicated in the trait, but for which there are suggested but incompletely explored links to the trait in the literature. These genes are represented by non-zero but low indirect support values, with the gene annotations with positive β values providing the specific prior links between the gene and the trait.

S.1.5.3 Gene set enrichment

Gene set enrichment is widely used to identify pathways or processes in which genes show stronger-than-expected associations within a genetic association study. For example, MAGMA regresses gene-level Z-scores on membership in a gene set. Gene sets with significant p-values (often, after FDR correction due to dependencies across gene sets) are considered to show enrichment. These enrichment p-values are analogous to marginal β values within the indirect support framework – that is, $\tilde{\beta}_j = \mathbb{E}\left[\beta_j | \mathcal{D}\right]$, where we marginalize over the latent variables d and β_{-j}. The marginal $\tilde{\beta}$ values give quantitative "effect sizes" for enrichment statistics, specifically the increase in log-odds of trait-relevance for genes in the gene set, ignoring all other gene sets besides gene set j. This marginalization step induces correlations across β values for overlapping gene sets, just as MAGMA enrichment results for overlapping gene sets are correlated.

S.1.5.4 Calibrating evidence sources

Many genes researchers study within experimental assays, including many potential drug targets, are chosen based on a hypothesis the researcher has regarding the role of gene in disease. These hypotheses often come from the literature, biomedical databases, or animal models. If these evidence sources are encoded as libraries of gene annotations, the marginal $\tilde{\beta}$ values for gene annotations within a library can be used to quantify the degree to which the corresponding evidence source is useful for prioritizing trait-relevant genes. Indirect support therefore both provides a quantitative justification for prioritizing genes based on prior observations, and it also enables calibration of the degree to which different evidence sources should be trusted.

S.1.5.5 Identifying independent disease mechanisms

While the marginal $\tilde{\beta}$ values estimate the degree to which a gene annotation in isolation affects a gene's probability of trait-relevance, the joint β values estimate the degree to which a gene annotation affects a gene's probability of trait-relevance after correcting for all other annotations. Therefore, when two gene annotations "overlap" (i.e., have many shared genes), at least one of

the joint β values will be significantly reduced relative to the marginal $\tilde{\beta}$ values. By contrast, two annotations that are independent (i.e., have few shared genes) will have joint β values not substantially different than their marginal $\tilde{\beta}$ values. The converse consequence of this property is that gene annotations with high joint β values indicate, to some extent, independent disease mechanisms (i.e., sets of genes likely to act through different biological pathways). This information can in principle be combined with other methods for partitioning genetic variants into independent mechanisms – for example, based on patterns of association across several traits[**?**] or epigenomic annotations[**?**].

S.1.5.6 Rare disease gene matching

For rare diseases, indirect support provides a principled way to prioritize genes that have evidence from the literature or databases of being involved in pathways relevant to a patient phenotype. One example of a widely used approach for prioritizing genes in this way is Exomiser, which (a) matches a patient phenotype (represented as terms in the Human Phenotype Ontology (HPO)) to a list of genes curated for the HPO terms; (b) propagates these to additional genes through semantic similarity to model-organism phenotypes or through protein interaction networks; and (c) uses various heuristics to combine these evidence sources into a score for each gene. An analogous approach using indirect support would be to (a) assign genes curated for the HPO terms to have high direct genetic support (i.e. probabilities of 95%, or some other value that accounts for the strength of curation evidence); (b) encode the animal model phenotypes, network interaction partners, and other information from the literature as gene annotations or gene sets; and (c) use PIGEAN to statistically infer which genes have the highest indirect support based on the input direct support values and the gene annotations. The benefit of the indirect support approach is that (a) it uses a statistical model with clear assumptions and a transparent outputs; and (b) it can handle a much larger variety of gene annotations given its regularization procedure and its ability to calibrate which annotations are disease-relevant using the input direct support values.

S.1.5.7 Analog with polygenic risk scores

The indirect support model is closely mathematically analogous to the polygenic risk score model initially proposed by LD-pred[**?**], which may aid in its interpretation. The goal of LD-pred is to infer the true effect sizes of SNPs on a phenotype, modeling a trait Y as a sum of the effect sizes of SNPs carried by an individual (encoded in a matrix X) in addition to an environmental noise term

$$Y = X\beta + \epsilon$$

LD-pred assumes that the marginal estimate of the SNP effects are available from univarate regressions of Y on each column of X, which is typically the case for GWAS summary statistics. It then converts these marginal $\tilde{\beta}$ estimates to joint SNP effect estimates, using a spike-and-slab prior.

This is the same model as we use for indirect genetic support, with gene annotations taking the place of SNPs and genes taking the place of individuals. Rather than a quantitative trait Y being influenced by SNP effects, we have a dichotomous variable for trait-relevance being influenced by annotation effects; this requires the linear regression coefficient estimates in the case of LD-pred to be replaced by logistic regression coefficient estimates in the case of PIGEAN. Another difference is that while in the case of LD-pred the trait Y values for individuals are observed, in the case of PIGEAN the trait-relevance values Y for genes are latent – we must therefore infer them from the observed association data. This final difference is why the annotation effect size inference procedure must be repeated within the outer Gibbs sampling loop for estimating the trait-relevance values; as these change, so too do the marginal and joint annotation effect estimates.

S.1.6 Burn-in and stopping criteria for the outer Gibbs sampler

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PIGEAN is fit using an outer Gibbs sampler that alternates between sampling gene-level trait relevance indicators d and updating annotation effects β (via the inner LDpred-style procedure), running multiple outer chains in parallel to assess convergence and reduce Monte Carlo error. After burn-in, additional outer iterations are run to obtain sufficiently precise posterior mean estimates of d and β.

Notation. Let $c \in \{1, \ldots, C\}$ index outer chains and t index outer iterations after burn-in. Let $\beta^{(c,t)} \in \mathbb{R}^M$ denote the vector of annotation effect sizes in chain c at iteration t, and let $p_i^{(c,t)} := \Pr(d_i = 1 \mid D, \beta^{(c,t)})$ denote the implied posterior probability for gene i (equivalently, the conditional mean of d_i given $\beta^{(c,t)}$ and the gene-level Bayes factor evidence). We use summaries across chains to monitor stationarity (burn-in) and Monte Carlo precision (post burn-in).

Active annotation subset (to avoid near-zero betas dominating diagnostics). Because the model uses a spike-and-slab prior, most β_j are near zero and can cause unstable or uninformative convergence checks (especially for relative error criteria). We therefore compute outer-loop diagnostics on an "active" subset of annotations, defined at diagnostic time as the L_β annotations with largest absolute posterior mean magnitude:

$$\mathcal{A} \;:=\; \{j \;:\; |\hat{\mu}_{\beta,j}| \text{ is among the top } L_\beta\}\,, \qquad \hat{\mu}_{\beta,j} \;=\; \frac{1}{C}\sum_{c=1}^{C} \bar{\beta}_{cj},$$

where $\bar{\beta}_{cj}$ is the within-chain mean of β_j over the post-burn-in draws accumulated so far. By default we use $L_\beta = 200$ (lenient preset) to focus diagnostics on the effects that can materially influence gene priors; under strict stopping L_β is increased to 500 to be more conservative.

Burn-in completion (stationarity). Burn-in ends when annotation effects are judged approximately stationary across chains. We compute $\widehat{R}_j$ (Gelman–Rubin $\widehat{R}$; as implemented in the codebase) for each β_j and summarize across the active subset $\mathcal{A}$ using a configurable quantile:

$$\widehat{R}_q^{(\beta)} \;=\; \text{Quantile}\left(\left\{\widehat{R}_j : j \in \mathcal{A}\right\}, q_{\text{burn}}\right).$$

Burn-in is declared complete once $\widehat{R}_q^{(\beta)} \le R_{\text{burn}}$ for S_{burn} consecutive diagnostic checks ("patience"), subject to minimum and maximum iteration constraints.

Defaults and justification. Monitoring $\max_j \widehat{R}_j$ over thousands of parameters is typically overly stringent; using an upper quantile (rather than a max) is substantially more robust in high dimension while still guarding against widespread non-convergence. We set lenient defaults intended for large-scale trait screening (where runtime is important) and a strict preset intended for final results.

Post-burn-in early stopping (Monte Carlo precision). After burn-in, the sampler continues until Monte Carlo standard errors (MCSEs) of posterior means are sufficiently small. We estimate MCSE using the variability of *chain means* across the C outer chains (this does not require storing full traces). For any scalar parameter θ (e.g. β_j or a gene probability p_i), define the chain mean

$$\bar{\theta}_c \;=\; \frac{1}{T}\sum_{t=1}^{T} \theta^{(c,t)}, \qquad \hat{\mu}_\theta \;=\; \frac{1}{C}\sum_{c=1}^{C} \bar{\theta}_c,$$

and the across-chain variance of chain means

$$s_\theta^2 \;=\; \frac{1}{C-1}\sum_{c=1}^{C}(\bar{\theta}_c - \hat{\mu}_\theta)^2.$$

We then estimate the MCSE of the overall mean $\hat{\mu}_\theta$ as

$$\widehat{\mathrm{MCSE}}(\hat{\mu}_\theta) \;=\; \sqrt{\frac{s_\theta^2}{C}}.$$

For annotation effects we use a relative MCSE (restricted to the active set):

$$\mathrm{relMCSE}_{\beta,j} \;=\; \frac{\widehat{\mathrm{MCSE}}(\hat{\mu}_{\beta,j})}{\max(|\hat{\mu}_{\beta,j}|, \epsilon_\beta)}, \qquad j \in \mathcal{A},$$

where ϵ_β is a small numerical floor to avoid division by values near zero. For gene-level probabilities we use an absolute MCSE and restrict attention to the top K genes by $\hat{\mu}_{p,i}$ (to avoid "max over all genes" pathology):

$$\mathcal{G}_K \;:=\; \{i : \hat{\mu}_{p,i} \text{ is among the top } K\}, \qquad \mathrm{MCSE}_{p,i} \;=\; \widehat{\mathrm{MCSE}}(\hat{\mu}_{p,i}),\; i \in \mathcal{G}_K.$$

Sampling stops when both criteria are met for S_{stop} consecutive diagnostic checks:

$$\mathrm{Quantile}\left(\{\mathrm{relMCSE}_{\beta,j}\}_{j\in\mathcal{A}}, q_{\mathrm{stop}}\right) \le \tau_\beta \quad \text{and} \quad \mathrm{Quantile}\left(\{\mathrm{MCSE}_{p,i}\}_{i\in\mathcal{G}_K}, q_{\mathrm{stop}}\right) \le \tau_p.$$

These checks are only evaluated after a minimum number of total outer iterations.

Command line configuration and defaults. Table 1 summarizes the outer-loop stopping flags. By default we use a lenient preset (appropriate for screening many traits) and provide a single switch, `--strict-stopping`, that applies a stricter preset (appropriate for final runs). Users can override any individual flag explicitly.

S.1.7 Optimizations for scaling PIGEAN to 7,000+ traits

The indirect support model requires two nested Gibbs sampling loops (an outer loop to sample the d values and an inner loop to sample the β values), a naïve implementation of which is computationally expensive. We made extensive optimizations to the PIGEAN algorithm so that it runs in under an hour on most traits and most gene annotation models, and in under ten minutes for typical traits and small gene annotation models.

First, all code is fully vectorized using numpy within Python. This includes the logistic regression step to compute the marginal $\tilde{\beta}$ values, the Gibbs sampling step to compute the joint β values, and the Gibbs sampling step to compute the d values. No loops are used in these steps (other than a loop over Gibbs iterations); everything is encoded via matrix multiplications, which effectively updates all annotation or gene values in parallel. Furthermore, because the vast majority of gene annotations are small and sparse, we use a sparse matrix representation of the gene annotation matrix X, which significantly reduces memory requirements and makes computations more efficient. Technically, in the Gibbs sampling step for computing β values, it is not possible to update all β values fully simultaneously, since correlated gene sets must be updated sequentially; we therefore (before beginning the Gibbs loop) batch gene sets into groups of relatively uncorrelated gene sets and then sequentially update each batch in parallel. Both the β and d Gibbs loops run multiple

Flag	Meaning	Lenient	Strict
`--strict-stopping`	If set, uses the strict preset for burn-in and precision thresholds (overridable by explicit flags).	off	on
`--r-threshold-burn-in` (R_{burn})	Burn-in $\widehat{R}$ threshold applied to $\widehat{R}_q^{(\beta)}$.	1.10	1.05
`--burn-in-rhat-quantile` (q_{burn})	Quantile of $\widehat{R}_j$ over active betas used to decide burn-in completion.	0.95	0.99
`--burn-in-patience` (S_{burn})	Number of consecutive diagnostic checks that must pass before burn-in is ended.	3	5
`--max-num-burn-in`	Optional hard cap on burn-in iterations (safety valve; burn-in cannot exceed `--max-num-iter`).	unset	unset
`--diag-every`	Frequency (in outer iterations) at which burn-in and stopping diagnostics are evaluated/logged.	5	2
`--stop-mcse-quantile` (q_{stop})	Quantile used when aggregating MCSE across monitored parameters for early stopping.	0.95	0.99
`--stop-patience` (S_{stop})	Number of consecutive diagnostic checks that must pass before sampling stops.	3	5
`--stop-top-gene-k` (K)	Number of highest-$\hat{\mu}_{p,i}$ genes used for gene-level MCSE monitoring.	200	500
`--max-abs-mcse-d` (τ_p)	Absolute MCSE threshold for $\hat{\mu}_{p,i}$ on the top-K genes.	0.02	0.01
`--max-rel-mcse-beta` (τ_β)	Relative MCSE threshold for $\hat{\mu}_{\beta,j}$ on active betas.	0.10	0.05

Table 1: Outer Gibbs burn-in and early stopping configuration. Lenient defaults are chosen to avoid pathological "max over thousands" behavior in high dimension and to yield practical runtimes in large-scale trait screens; strict defaults provide more conservative stationarity and precision targets for final runs.

chains to monitor convergence and reduce sampling variance; we update all chains in parallel using vectorization and 2-D or 3-D tensors (for the outer d loop, the tensor is gene values by chains, and for the inner β loop, the tensor is gene annotation values by outer chain by inner chain).

Second, we discard gene annotations from the Gibbs loop that are unlikely to have non-zero effect sizes; this introduces an approximation but with reductions in computation of several orders of magnitude (as well as sparser and easier to interpret outputs). Before any Gibbs sampling is conducted, we apply several filters:

- If the p-value of the marginal logistic regression of the initial HuGER scores on the gene annotation is above a threshold (default $p = 0.01$) we ignore the annotation.
- If the marginal $\tilde{\beta}$ for the gene annotation (according to the initial HuGER scores) is negative, we ignore the annotation. Empirically, for reasons discussed below (**??**), most gene annotations with negative $\tilde{\beta}$ values are set to zero by the spike and slab prior.
- We restrict the Gibbs loop to a maximum of 5000 gene annotations (this number is configurable). We select these gene annotations according to a greedy method that iteratively adds the gene annotation with the highest joint β value (as inferred from the initial HuGER scores and the initial $\tilde{\beta}$ values), then ignores gene annotations correlated with that annotation, and then repeats the process.
- We prune gene annotations by removing those with pairwise correlation above $r^2 = 0.8$. We calculated both unweighted correlations and correlations weighted by the initial β values.
- Prior to each iteration of Gibbs sampling, before running the logistic regression procedure to

calculate the marginal $\tilde{\beta}$ values, we run a faster linear regression procedure. We ignore (from that iteration of Gibbs sampling) any gene annotations with $p > 0.01$. This allows us to limit the more costly logistic regression procedure to a smaller number of gene sets.

- We set β values to zero if they are either below 0.001 or below $0.01 \max \beta$. This encourages a sparse solution which makes the model more stable and also more computationally efficient.

S.1.8 Confounder correction within HuGER scores

By their nature, HuGER scores are biased towards genes that have many SNPs linked to them and/or are "island" genes and far from other genes. The impact of confounders on gene-level association scores is not unique to HuGER scores and accounted for in most gene set enrichment approaches[]. For HuGER scores, we compute the following gene-level confounders:

- The number of genes within 250kb upstream of the gene
- The number of genes within 250kb downstream of the gene
- The sum of the reciprocal of distances to all other genes
- The length of the gene
- The number of SNPs linked to the gene when SNPs are randomly distributed across the genome.

We then regress the HuGER scores on these confounders, and take the residuals as the adjusted HuGER scores that we use within PIGEAN

Empirically, because this correction procedure is not perfect, additional confounders can also occur at the gene annotation level. For example, the `LAKE_ADULT_KIDNEY`[] gene annotation in MSigDB[] appears to be enriched for island genes. We therefore apply a further correction procedure where we adjust the gene annotation $\tilde{\beta}$ Z-scores by regressing them on the average confounder value for genes with the annotation. We then adjust the standard error of the $\tilde{\beta}$ value so that the Z-score equals the residual from the confounder regression.

S.1.9 Internal parameters and default choices used in HuGER scores

Parameters for the HuGER method include parameters for mapping SNP p-values to posterior probabilities of association, identifying independent signals (i.e. clumps of SNPs), linking SNPs to genes, and normalizing gene posteriors within a region. All of these parameters are configurable within the PIGEAN software package.

Parameters for map SNP p-values to posterior probabilities of association specify the prior allelic variance of SNPs[] and the prior probability of association for a random SNP in the genome. Since the scale of effect size units is often unknown for GWAS summary statistics, and the prior allelic variance may vary by trait, we instead constrain these two parameters by specifying the posterior probabilities of association at two SNP p-value thresholds, and then adjust these for the number of associations produced by the GWAS to correct for the dependence on sample size that this approach induces[]. The default values used by PIGEAN are a posterior probability of 0.75 for $p = 5 \times 10^{-8}$ and a posterior of 0.01 for $p = 0.01$. When there are more than 100 independent signals with $p < 5 \times 10^{-8}$, we update the first p-value threshold to be the p-value of the hundredth strongest association. One final parameter used in this step is the minimum sample size required for a SNP to be included in analysis; if the sample size of the SNP (or inverse of standard error

squared if sample sizes are unavailable) is less than 0.5 times the average sample size in the GWAS, then we exclude it from analysis.

Parameters for identifying independent signals are motivated by our desire for the method to be fast and easy to apply without the need for assembling auxiliary files beyond GWAS summary statistics. We therefore elected to conduct SNP clumping without access to LD values between SNPs (extending PIGEAN to use LD would be trivial and could be done in several ways, including applying the PLINK clumping procedure[] or any other fine mapping procedure such as COJO[] or FINEMAP[]). To clump SNPs into signals, we iteratively identify an index SNP, which is the SNP with the lowest p-value among all remaining SNPs, and we then remove from further consideration all variants within a contiguous region of the SNP. The contiguous region is obtained as follows:

- It must be at least 250kb both upstream and downstream of the index SNP.
- For each side, either all SNPs within the final 100kb must have $p > 10^{-5}$, or the first SNP i outside of the region must have $\log(p_i) > 1.3 \times \overline{\log(p)}$, where $\overline{\log(p)}$ is the average $\log(p)$ for SNPs in the region. This is heuristic to identify independent signals without LD, as p-values of SNPs in LD with the index SNP typically become less significant as a function of distance

Parameters for linking SNPs to genes are used only for non-coding SNPs without prespecified SNP-to-gene linkage scores[]. If an auxiliary file is specified with direct SNP-to-gene probabilities, these are used to assign SNPs to genes. If the SNP is in an exon of the gene, it is assigned a probability of 95% of being linked to the gene. Otherwise, we use a logistic decay function to calculate linkage probabilities. To fit this function, we randomly sample SNPs across the genome, and we then convert these to labeled (distance, 0/1) pairs by (a) calculating the distance to the closest gene and (b) assigning a 1 with probability of 70% and 0 otherwise. We then fit a logistic regression model to map distance from a gene to probability that the SNP is linked to it. The function we use by default is $\log\left(\frac{l}{1-l}\right) = -5.086 \times 10^{-5}x + 2.988$, where l is the probability that the SNP is linked to the gene and x is the distance from the SNP to the gene. We note that the logistic function is used primarily for convenience, and a Weibull distribution may be a better parametric fit[].

The final set of HuGER parameters normalize gene probabilities of trait-relevance relative to each independent signal. We require that, after running the HuGER method to calculate d_i values for each $\mathcal{D}_k$ in Equation (S2),

$$\sum_i \Pr\left(d_i|\mathcal{D}_k\right) = 1$$

This is the assumption that there is exactly one causal gene per signal, and it sometimes requires scaling the SNP to gene linkage probabilities to be higher than predicted by the input values or logistic regression (i.e. if the signal lies far from every gene). This can be disabled within PIGEAN, which is equivalent to assuming that there is something other than an annotated gene that mediates the association signal. We also restrict the maximum probability $\Pr\left(d_i|\mathcal{D}_k\right)$ to be below 0.9, to avoid overconfidence of the model.

S.1.10 Alpha regularization

A long appreciated issue with methods that use gene annotations to prioritize disease-relevant genes is that they are biased toward well-studied genes. In our model, this is manifested as a bias toward genes with many annotations (or, equivalent, a high variance across values for its row in the X matrix). It is worth noting that, when annotation effect sizes are calibrated from observed data, this feature is not as obvious as it might at first appear – the β values have prior expectations

of zero, and if their posterior expectations were also zero, then there would be no bias of higher d values toward well studied genes. However, for reasons we explore through simulationsS.1.12, the posterior β values are biased toward positive values for several reasons. Therefore, absent regularization, well-studied genes will in fact have higher d values (this feature is also an empirical property of PoPS, for which our analysis of PoPS as an approximation to indirect provides an explanation).

Our use of the regularization term α therefore corrects for this bias. We fit α to guarantee that the posterior expectation of d_i is independent of the number of annotations for gene i. Therefore, indirect support values produced by PIGEAN are not biased toward well-studied genes. Our simulations show that this improves accuracy of the indirect support inferences (Supplementary Figure **??**). However, it is possible that it may be desirable in some cases to not correct for how well-studied a gene is in indirect support calculations, and this topic is worthy of further research in the future.

S.1.11 Properties of the PIGEAN algorithm as demonstrated through simulations

S.1.12 Validation on Simulations

As a negative control, we first applied PIGEAN to the 20 "null" simulated GWAS and found that, as expected, inferred gene annotation effect sizes were all close to zero (Extended Data Figure **??**a) as were indirect, direct, and combined gene ABFs (Supplementary Figure **??**bcd). Similarly, when we applied PIGEAN to the 50 real GWAS but randomly permuted genes within the 166,868 gene annotations, we also observed annotation effect sizes close to zero (Supplementary Figure **??**e). Across our 20 null simulations and 166,868 annotations only one gene had combined gene ABF above 3 (Supplementary Figure **??**g), with the dominant contribution coming from direct ABFs (Supplementary Figure **??**h), and across the 50 real GWAS with permuted annotations, only one annotation had effect size above 0.004 (Supplementary Figure **??**f). Based on these results, annotation effect sizes of 0.005 and gene combined ABFs of 3 roughly correspond to "study-wide significant" p-value thresholds (Supplementary Table 3).

We also used simulations with non-zero annotation effect sizes to understand the behavior of PIGEAN in response to different model parameters. Accuracy of PIGEAN is affected by noise from two sources: the incomplete ability of annotations to predict disease-relevant genes ("annotation noise"), and the incomplete ability of summary statistics to identify disease-relevant genes ("association noise"). Focusing first on annotation noise, we ran PIGEAN with a truncated model in which we treated each gene's probability of disease relevance (d_i) as observed rather than inferred from summary statistics. These simulations yielded four main findings. First, PIGEAN in general conservatively estimates both annotation effect sizes (Supplementary Figure **??**a) and gene ABFs (Supplementary Figure **??**b), due to its "spike and slab" regularization that sets estimated effect sizes to zero and shrinks estimated effect sizes toward zero. Second, annotations with negative effects (i.e. that reduce genetic support) were shrunk substantially more than those with positive effects, as (consequently) were genes with ABFs ¡ 1, an asymmetry that naturally results from the higher standard errors for negative gene set effect sizes when there are only a small number of disease-relevant genes (Supplementary Figure **??**cd). Third, PIGEAN's accuracy was also affected by simulation parameters, with more accurate inferences possible for larger simulated effect size magnitudes (larger σ Supplementary Figure **??**a) and (to a lesser extent) more non-zero simulated effect sizes (larger p; Supplementary Figure **??**b). Additional noise added to the simulated gene ABFs affected accuracy of the inferred ABFs but (perhaps non-intuitively) did not affect accuracy

of the inferred annotation effect sizes (Supplementary Figure **??**c). Finally, due to the nature of Gibbs sampling, estimates produced by PIGEAN varied across runs, but with sufficient numbers of iterations and chains estimates were stable (Supplementary Figure **??**).

These simulations confirmed the importance PIGEAN's joint inference and regularization features. First, its full joint inference procedure (in which gene and gene set values are iteratively re-estimated) increased accuracy compared to a model in which gene set effect sizes were only estimated once (Supplementary Figure **??**). Second, even though simulated gene ABFs had no dependence on the number of annotations for the gene, regularization (via its parameter) to penalize genes with many annotations was necessary to avoid biases and ABF accuracy reductions induced (as described in Supplementary Figure **??**cd) by asymmetric annotation standard errors (Supplementary Figure **??**). Similarly, even though simulated annotation effect sizes had no dependence on the number of genes with the annotation, regularization (via its w parameter) to penalize "large" annotations was necessary to avoid biases induced by the smaller standard errors of larger annotations (Supplementary Figure **??**; Methods).

For simulations with a small number of non-zero annotation effects, PIGEAN was able to infer reasonably accurate estimates of its sparsity parameter (p) (Supplementary Figure **??**), although it significantly underestimated p as the fraction of non-zero effects increased. Our attempts to estimate PIGEAN's shrinkage parameter (σ) for individual simulations suggested that there are typically too few annotations to estimate this parameter accurately, and we therefore tuned this parameter empirically (Methods). Fortunately, varying these parameters from their simulated values by even an order of magnitude did not dramatically reduce PIGEAN's estimation accuracy (Supplementary Figures **??** and **??**).

In addition to these simulations focused exclusively on "annotation noise", we also evaluated the impact of "association noise" on PIGEAN's performance – that is, the reality that (due to finite sample sizes and the vagaries of evolution) many disease-relevant genes will not lie near detectable association signals. We studied association noise in the context of GWAS summary statistics, given the well-documented difficulties in inferring effector genes from these data. For each of three annotation noise settings (Methods), we simulated five GWAS of increasing sample sizes (using linkage disequilibrium from the 1000 Genomes Project, SNP to gene linkages from a previously published method, and SNP effect size parameters from previously published simulations; Methods). As expected, the use of GWAS data rather than observed disease probabilities degraded the accuracy of PIGEAN (Supplementary Figure **??**a). With perfect knowledge of SNP to gene linkages Supplementary Figure **??**b), less annotation noise Supplementary Figure **??**c), or larger sample sizes Supplementary Figure **??**d), the accuracy of PIGEAN increased, but only to limited extent. Accuracy was much higher for marginal annotation effect sizes than for joint effect sizes Supplementary Figure **??**e), which is a common feature of multivariate models with correlated predictors. When PIGEAN inferred its regularization parameters, rather than being given the true parameters used in simulations, accuracy degraded further Supplementary Figure **??**f). Notably, however, in all scenarios PIGEAN conservatively estimated annotation effect sizes, suggesting that the overwhelming impact of GWAS annotation noise is to decrease power rather than to increase false positives.

We conclude that, under idealized simulations, PIGEAN behaves broadly as expected. Three noteworthy properties are (a) the difficulty of inferring negative genetic support (due to known properties of GWAS but perhaps unappreciated properties of models with few disease-relevant genes); (b) the importance of regularization and iterative inference in gene set analysis, and (c) it conservative nature when confronted with annotation noise and and association noise.